Project Summary: Manpower Prediction via Order Forecasting

1. Project Overview
Developed an end-to-end predictive analytics model to forecast manpower requirements by first predicting future order volumes. The objective was to synchronize staffing levels with e-commerce demand fluctuations, ensuring operational efficiency across warehouses and fulfillment centers.

2. Data Integration & Sources (Multi-ERP Architecture)
Extracted and consolidated high-volume data from multiple enterprise systems:

- **ERP Systems:** Integrated data from **SAP ERP**, **Unicommerce**, and **eShopBox** to get a unified view of inventory and order flow.
- **Order History:** Analyzed historical sales and order data to identify growth patterns and daily/weekly trends.
- **Operational Metrics:** Collected data on processing time per order (Pick-Pack-Ship) to link order volume with labor hours.

3. Technical Approach

- **Two-Stage Modeling:**
  - **Stage 1 (Demand Forecasting):** Used **Time Series Analysis** (Prophet/XGBoost) on historical order data from ERPs to predict upcoming order spikes.
  - **Stage 2 (Manpower Mapping):** Converted predicted order volumes into required man-hours using productivity coefficients (e.g., Orders per Head per Shift).
- **Preprocessing:** Handled data inconsistencies across different ERP formats and performed feature engineering for seasonality (e.g., Sale events, Festivals).
- **Validation:** Monitored prediction accuracy using **MAE** and **RMSE** to ensure the workforce stayed within +/- 5% of actual demand.

4. Business Impact & Results

- **Data Consolidation:** Successfully bridged data silos between SAP, Unicommerce, and eShopBox for a "single source of truth."
- **Proactive Staffing:** Improved the ability to predict manpower gaps 2-4 weeks in advance, reducing reliance on expensive last-minute temporary labor.
- **Efficiency:** Achieved a **15-20% reduction in labor overheads** by optimizing shift schedules based on predicted order density.

2- Project Summary: Multi-Channel Inventory & Fulfillment Optimization

1. Project Overview
Developed an integrated analytics solution to synchronize inventory levels and fulfillment across multiple platforms. By consolidating data from **SAP, Unicommerce, and eShopBox**,

the project aimed to reduce 'Stock-outs' and 'Overstocking' while improving the Order-to-Dispatch cycle time.

2. Data Integration (Unified ERP Strategy)

- **SAP ERP:** Extracted core procurement and master inventory data.
- **Unicommerce & eShopBox:** Pulled real-time order flow and marketplace fulfillment data.
- **Historical Orders:** Analyzed 2+ years of order history to identify fast-moving vs. slow-moving SKUs (Stock Keeping Units).

3. Technical Approach

- **Demand Forecasting:** Applied **XGBoost** and **Random Forest** models to historical order data to predict future SKU-level demand.
- **Inventory Health Analysis:** Categorized inventory using **ABC/XYZ analysis** based on value and volatility derived from ERP data.
- **Automation:** Created a data pipeline to merge disparate data formats from different ERPs into a single Python-based analytical dashboard.
- **Metrics:** Focused on reducing **Dead Stock** and improving **Inventory Turnover Ratio**.

4. Business Impact & Results

- **Reduced Lost Sales:** Decreased stock-out instances by **12%** during peak sale events by predicting high-demand items.
- **Warehouse Efficiency:** Optimized the 'Pick-Path' in the warehouse by placing high-velocity items (predicted by the model) in accessible zones.
- **Working Capital Optimization:** Reduced stagnant inventory by **18%**, freeing up significant business capital.